Energy Conversion in a Resonant Cavity

From Wind to Electricity

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Thematic positioning: Physics (Mechanics) · Physics (Matter) · Computer Science

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Outline

1. Context & Motivation
2. Research Question & Objectives
3. Theoretical Model
4. Experimental Setup
5. Results
6. Real-World Conditions
7. Conclusion & Perspectives

Energy Context

Challenges of the energy transition

- Diversification of renewable energy sources
- Conversion at **every scale**: wind turbines → piezoelectric devices
- Need for autonomous sensors in remote areas

Origin of the inspiration

The powerful vibrations of **wind instruments** in chamber music led to the idea of transposing their physical principles towards electricity generation.

Solar energy
↓
Wind energy
↓
Acoustic vibrations
↓
Electricity

Helmholtz Resonators – Examples & Context

Mechanical energy sources (wind)

- *Onshore/offshore* wind turbines
- Kite power generators
- Wave energy converters

Acoustic resonance – applications

- Acoustic guitar
- MRI (medical imaging)
- Car exhaust line

Types of Helmholtz resonators

- Series resonators
- Acoustic metasurfaces
- Hilbert fractal resonators

Key observation

These systems amplify or attenuate specific frequencies – can we **extract electrical energy** from them?

Research Question & Objectives

Research question

Is there an alternative method to harness
the mechanical energy of wind?

Objectives

1. Model and verify the **natural frequency** of a Helmholtz cavity
2. Characterise the **coupled resonance** between a vibrating blade and the cavity
3. Measure the **voltage produced** by a piezoelectric quartz
4. Evaluate **performance** under real-world conditions
5. Explore **perspectives**: autonomous IoT sensors

Keywords

Helmholtz resonance
Coupled resonance
Piezoelectricity
Energy conversion

Working hypothesis

Constant airflow (steady wind) $\Rightarrow$ forced oscillator at a fixed frequency.

Natural Frequency of a Helmholtz Cavity

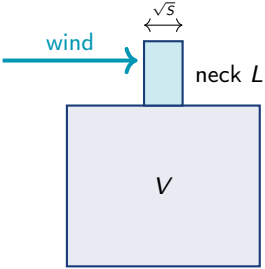
Equivalent mechanical model

- Air mass in the neck $\leftrightarrow$ mass
- Cavity volume $\leftrightarrow$ spring

Resonant frequency

$$f_0 = \frac{c}{2\pi}\sqrt{\frac{S}{VL}}$$

c = speed of sound, S = neck cross-section
V = cavity volume, L = effective neck length



Numerical application

- 33 cl can: $f_0^{\text{theo}} = 335\,\text{Hz}$
- 50 cl can: $f_0^{\text{theo}} = 256\,\text{Hz}$

Derivation of f_0

Pressure–density relation

$$p_i = \frac{c^2 Q}{j\omega V} \qquad (1)$$

Newton's 2nd law (neck, $m = \rho_0 S L$)

$$j\omega L\, Q = S(p_0 - p_i) \qquad (2)$$

Substituting (1) into (2):
p_0 minimum $\Longrightarrow$ resonance:

$$\omega_0^2 = \frac{c^2 S}{VL} \Longrightarrow \boxed{f_0 = \frac{c}{2\pi}\sqrt{\frac{S}{VL}}}$$

Key parameter: quality factor Q

$$Q = \frac{f_e}{\Delta f}$$

Frequency sweep – 33 cl: $Q \approx 20$ Free oscillation – 33 cl: $Q \approx 100$

Coupled Resonance: Vibrating Blade + Cavity

Coupled system (mechanical oscillators)

$$\begin{cases} m_p\ddot{x}_p + R_p\dot{x}_p + (k_p+\mu A^2)x_p + \mu AS\, x_a = F(t) \\ m_a\ddot{x}_a + R_a\dot{x}_a + \mu S^2 x_a + \mu AS\, x_p = 0 \end{cases}$$

Two coupled natural frequencies:

$$\omega_\pm^2 = \frac{\omega_p^2+\omega_h^2}{2} \pm \frac{1}{2}\sqrt{(\omega_p^2-\omega_h^2)^2 + 4k^2\omega_p^2\omega_h^2}$$

Coupling coefficient

$$k^2 = \frac{\Delta^2 - \frac{1}{4}(\omega_p^2 - \omega_h^2)^2}{\omega_p^2 \cdot \omega_h^2}$$

	33 cl	50 cl
k	≈ 1.10	≈ 1.25

Very strong coupling in both cases.

Devices & Protocols

Helmholtz resonators

- Aluminium cans: 33 cl and 50 cl
- Neck = top opening of the can

Transducers

- Microphone (acoustic measurement)
- Piezoelectric quartz (mechanical → electrical)
- Digital acquisition unit

Three experimental protocols

1. **Frequency sweep** (speaker + microphone): verification of f_0, measurement of Q
2. **Free oscillation** (ball + microphone): mechanical impulse, measurement of Q
3. **Coupled resonance** (speaker + piezo): observation of both modes

Unsuccessful attempts

Vibrating blade, speaker membrane, tuning fork, ball, rod, plastic – only the piezoelectric quartz proved conclusive.

Acquisition Chain & Signal Processing

Digital processing (Python)

- Low-pass filter: discrete integration
- FFT spectrum, automatic peak detection

Power measurement

$$P = \frac{U^2}{R}, \qquad E = \int P \, \mathrm{d}t$$

Impedance matching: optimal load resistance R.

Natural Frequencies: Theory vs. Experiment

Can	f_0^{theo}	f_0^{sweep}	f_0^{breath}	$f_0^{\text{free osc.}}$	Deviation
33 cl	335 Hz	301 Hz	345 Hz	300 Hz	$\leq 4\,\%$
50 cl	256 Hz	245 Hz	–	250 Hz	$\leq 4\,\%$

Quality factor

- Sweep – 33 cl: $Q \approx 20$
- Sweep – 50 cl: $Q \approx 10$
- Free oscillation – 33 cl: $Q \approx 100$

Discrepancy between methods: radiation losses during forced sweep.

Coupled resonance (50 cl)

- Lower mode: 265 Hz (predicted 256 Hz)
- Upper mode: 480 Hz

Coupled resonance (33 cl)

- Lower mode: 340 Hz
- Upper mode: 580 Hz

Piezoelectric Quartz Response & Extracted Power

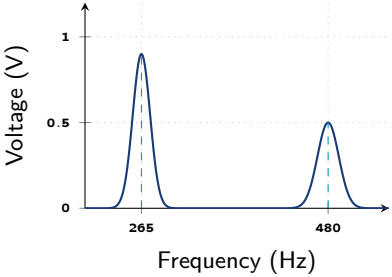
Key result

- Maximum voltage at $f = 256\,\text{Hz}$ (50 cl can)
- Sharp peak at the Helmholtz resonance frequency
- Impedance matching: maximisation of $P = U^2/R$

Extracted power (per can)

$$P_{\text{can}} = 3 \times 10^{-6}\,\text{W} = 3\,\mu\text{W}$$

Two voltage peaks observed: Helmholtz mode + upper coupled mode.



Schematic spectrum – 50 cl can

Tests Under Real-World Conditions

Protocol

- Cans exposed to a **natural outdoor airflow**
- Simultaneous measurement: microphone + piezoelectric
- Recording on the acquisition unit

Observations

- Excitation of natural modes detected
- Piezoelectric signal measurable even at low airflow
- Results consistent with laboratory measurements

Hypothesis validation

- Constant flow ✓ steady wind
- Linearity ✓ small amplitudes
- Thermal independence $\approx$ at ambient T

Effect of water (50 cl)

Adding water $\Rightarrow$ change in effective volume $\Rightarrow$ shift in f_0.

Power Budget

Reference	Power	Ratio
Single can (this study)	$3\,\mu$W	$\times 1$
Solar cell (calculator)	$\sim 0.1\,\mu$W	$10\times$ less
Microcontroller (sleep mode)	~ 1–$5\,\mu$W	comparable
Phone charger	~ 5 W	$1.6\times 10^6\times$ more
LED bulb	~ 10 W	$3\times 10^6\times$ more

Scalability

To power a 45 W portable cooler:

$$N = \frac{45}{3\times 10^{-6}} = 15\times 10^6 \text{ cans}$$

Total area: $A \approx 6\times 10^3\,\text{m}^2$ (0.6 ha)

LoRa application

Energy per transmission: $E \approx 0.01$ J

Charging time (1 can):

$$t = \frac{0.01}{3\times 10^{-6}} \approx 55\,\text{min}$$

Range: $D_{\text{max}} \approx 50$ km

Application Perspectives

Compatible sensors (power requirements)

Sensor	Power
Gas	$\sim 1\,\mathrm{mW}$
Pressure	$\sim 10\,\mu\mathrm{W}$
Humidity	$\sim 5\,\mu\mathrm{W}$
Temperature	$\sim 1\,\mu\mathrm{W}$

Temperature and humidity sensors are directly reachable with a small network of cans.

Potential deployment sites

- Air quality monitoring (coastlines)
- Avalanche risk & weather (mountain passes)
- Wildfire detection (forests)
- Pollution / vibrations (bridges, rooftops)

Improvement directions

- Advanced piezo materials (PVDF)
- Impedance coupling optimisation
- Resonator arrays (metasurfaces)

Conclusion

Summary

- **Validated model:** $f_0^{\text{theo}} \approx f_0^{\text{exp}}$ within 4 %
- **Quality factor** measured by two independent methods
- **Strong coupling** confirmed: $k \approx 1.1$–1.25
- **Measured power:** $P = 3\,\mu\text{W}$ per can
- **Real-world tests:** coherent signal under natural wind

Answer to the research question

A Helmholtz cavity coupled with a piezoelectric quartz constitutes a viable alternative method for converting mechanical wind energy, at a scale suited to powering **autonomous IoT sensors**.

Thank you for your attention

Denis O – TIPE 2024–2025

Appendix A – Full Derivation

1. **Pressure–density relation:** $p_i = c^2 \rho'$, where $c^2 = \partial P / \partial \rho|_S$
2. **Mass conservation:** $Q = V\, \partial \rho / \partial t \;\Rightarrow\; p_i = c^2 Q / (j\omega V)$ (1)
3. **Newton's law (neck):** $j\omega L\, Q = S(p_0 - p_i)$ (2)
4. **Substituting (1) into (2):** p_0 minimum $\Leftrightarrow \omega^2 = c^2 S / (VL)$

$$\boxed{f_0 = \frac{c}{2\pi}\sqrt{\frac{S}{VL}}}$$

Appendix B – Coupling Summary Table

		33 cl	**50 cl**
Blade alone	f_p	301 Hz	250 Hz
Cavity alone	f_h	335 Hz	256 Hz
Lower coupled mode	f_-	340 Hz	265 Hz
Upper coupled mode	f_+	580 Hz	480 Hz
Coupling coefficient	k	≈ 1.10	≈ 1.25

Appendix C – References

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